Engineering Ethics Assignment for discussion in Week 10

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Week 7:

Give your idea on how we can manage the problem of increasing dishonesty among scientists. (300 words)

Increasing dishonesty in scientists is caused by several reasons, including conflict of interest, publication pressure, and/or motivation for reward from publishing statistically significant research with results rather than less reward for publishing research with null or statistically insignificant results.

A scientific article "Dishonesty in Scientific research" written by Nina Mazar and Dan Ariely in the Journal of Clinical Organization, which is published by The American Society for Clinical Investigation, revealed that people may be dishonest given a conflict-of-interest situation. "Ethics research, including our own work, has shown that dishonesty is not about good versus bad people, rather it is primarily about conflicts of interest and motivated reasoning". Dishonest actions are all too prevalent in everyone's daily lives, because people are constantly encountering situations in which they can gain advantages by cutting corners.

The article also stated the academic research system and its reward system contributes to the dishonesty of academia. The academic research system rewards statistically significant research findings with prestigious publications, grants, and promotions. Statistically nonsignificant research findings, on the other hand, are almost entirely disregarded, even though we sometimes learn more from them. Consequently, the system sets up a conflict of interest when, after thousands of dollars of research funding and hundreds of hours of work, one faces null effects. This would likely to motivate people to overreport positive results and underreport negative data, to disregard data that do not fit or to selectively show results that validate the model, even if they do not represent a typical outcome. With limited research dollars, pressure to produce positive results, and increasingly in-depth studies, the opportunity exists to rationalize biased presentation of data.

In the Engineering Ethics course, we learned several real-life cases of (possible) dishonesty in science: several scientists, including Peter Daszak, wrote that ""We stand together to strongly condemn conspiracy theories suggesting that COVID-19 does not have a natural origin." and declared that there was no conflict-of-interest

in their article “Statement in support of the scientists, public health professionals, and medical professionals of China combatting COVID-19” in Lancet. However, Peter Daszak, one of the author of the latter is the president of the EcoHealth Alliance, which send some of the research fund awarded by the NIH to the Wuhan Institute of Virology. Thus, this is a form of conflict of interest. Another one was when several scientists including Kristian Andersen wrote that the SARS-CoV-2 virus is not a laboratory construct or a purposefully manipulated result

Several ways to combat the increasing dishonesty among scientists are to:

1. Preventing scientists from conducting and publishing results that have conflict of interest – publishing articles that are related to a company that funded the research. For example, Peter Daszak that was funded by the Wuhan Institute of Virology
2. Reward scientists and articles/journals that are done in an honest way even with null results or statistically insignificant results.
3. Impose sanctions and penalties if scientists are caught lying or publishing false results
4. Have the research/ article/journal reviewed by an independent body or organization to
5. An interesting finding in the article “Dishonesty in Scientific research” written by Nina Mazar and Dan Ariely showed that if there is a honor code or moral code in the beginning of a trial or research, it may decrease dishonesty done by the experimenters. “having to read aloud and sign an honor code just before indicating the procedure performed on a patient for health insurance claim purposes may deter some doctors from faulty reporting”
6. Having ethics and honesty taught to students in education.

Week 8:

What kind of restriction, oversight, and penalty do you think is needed to control the risk of gain of function research? (300 words)

Oversight and Regulation

1. Multi-layered review process: GOF research proposals should undergo rigorous evaluation by multiple bodies, including funding agencies, institutional biosafety committees, and specialized review board

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2. Case-by-case assessment: Each GOF experiment should be evaluated individually, considering its scientific merit, potential risks, and whether less risky alternatives exist
3. Ongoing monitoring: Continuous oversight throughout the lifecycle of GOF experiments is crucial to ensure compliance with safety protocols and to address any emerging concerns
4. International cooperation: Establishing globally accepted definitions and standards for GOF research can help ensure consistent safety measures across different countries

Risk Mitigation Strategies

1. Enhanced biosafety measures: Implement stringent biosafety protocols in laboratories conducting GOF research to minimize the risk of accidental release
2. Biosecurity measures: Develop robust systems to prevent potential misuse or theft of enhanced pathogens
3. Transparency: Increase public disclosure of GOF research activities and findings to foster trust and enable broader scientific scrutiny

Penalties and Enforcement

1. Funding revocation: Institutions that fail to adhere to federal guidelines for genetic engineering risk losing their funding entirely
2. Legal consequences: Researchers and institutions violating established protocols could face legal action, especially if their negligence leads to biosafety incidents.
3. Professional sanctions: Scientists engaging in unauthorized or improperly conducted GOF research might face professional repercussions, including loss of research privileges or damage to their reputation.

Week 9:

What measures would you take to realize inspections of lab activities to see whether the Omicron variant is a lab construct or not? (300 words)

To see whether the Omicron variant is a lab construct or not, first made a group of compare the the genetic sequence of the Omicron variant and compare the group of mutations to previously known SARS-Cov mutations.

The High Wire reports the finding from two Japanese scientists Atsuki Tanaka and Takayuki Miyazawa that the variant Omicron has different mutations than other variants- its mutations is not built on previous variants of the virus like the evolutionary method, but is different. Tis suggests that Omicron is possibly a lab construct.

Other measures might include,

1. Genomic analysis: Conduct detailed sequencing and comparison of Omicron against known natural and engineered coronavirus strains to identify any unusual patterns.
2. Mutation spectrum analysis: Compare Omicron's mutation patterns to those associated with natural evolution versus laboratory settings.
3. Phylogenetic studies: Trace Omicron's evolutionary history to determine if it fits natural patterns.
4. Lab records review: Inspect research facility records for projects involving similar genetic sequences or mutation patterns.
5. Equipment and materials audit: Examine lab equipment and materials used in coronavirus research.
6. Biosafety protocol assessment: Review containment measures in relevant facilities.
7. Animal model investigations: Focus on animal models used in research, particularly mice.
8. Collaborative international inspections: Implement coordinated inspections across multiple countries.
9. Bioinformatics analysis: Use advanced computational tools to analyze Omicron's spike protein structure and function.
10. Expert panel review: Assemble independent experts to evaluate all collected data.